Affan School Management System

SOFTWARE DESIGN DESCRIPTION DOCUMENT

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BS IT

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Project Title: Affan School Management System

Revision History

Date	Description	Author	Comments
date	Version 1.0	Your Name	First Revision
		Mehran Jamil	

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	Mr. Faisal Shahzad	Supervisor DIT	10-01-2026

Affan School Management System

Introduction

Affan School Management Systems_Plays an essential role in the current educational system. School authorities all over the world are engaged in a lot of day-to-day administrative and academic activities to manage and provide a better academic experience to students effectively. However, maintaining and keeping track of school administrative activities is not an easy process in the fast-growing world. It requires hard work and often it is time-consuming.

School ERP is a web based School Management System that enables school to use and operate many of integrated interrelated modules and manage the administration of school efficiently.

We offer School ERP which simplifies and automates schools administration process. The School Management System is accurate and reliable and can be conveniently accessed from school intranet as well as from the public internet. It is fully browser- based school management system which also includes virtual campus which can be linked with school portal and contains powerful online access to bring parents, teachers and students on a common interactive platform.

1.1 Purpose

The parents of students are very busy now days, so they can't monitoring their children and them activities a properly and regular. This school management system helps the parents monitor their children from anywhere. They can check their children's academic performance from a remote location.

1.2 Scope

E-School Management System is comprehensive web-based School Management Software. It will design for better interaction between students, teachers, parents & management.

1.3 Definitions, Acronyms, and Abbreviations.

This document is formed using IEEE SRS format, headings are in bold capital letters and wherever necessary diagram is provided.

- SMS - school management system
- OSMS - Online school Management System
- PPEE - Php Platform Enterprise Edition
- SQL - sql Server Page
- SRS - Software Requirements Specification
- OS - Operating System

1.4 References

https://www.academia.edu/24648944/SOFTWARE_REQUIREMENTS_SPECIFICATION_SR_FOR_STUDENT_INFORMATION_MANAGEMENT_SYSTEM

- Software Engineering a Practitioner's Approach by Roger S. Pressman.
- Class Power Point presentation slides.
- For Database sql.

1.5 Overview

School Management System can provide following benefits in general to your school, college or institute:

- Single system to manage all School related information from anywhere in the School.
- Easy to use school ERP
- Reliable and secure system Complete
- Automation of operations More Time to focus on Strategic Tasks
- Better informed decision making for management
- Parents have access to all academic information about their wards through the internet

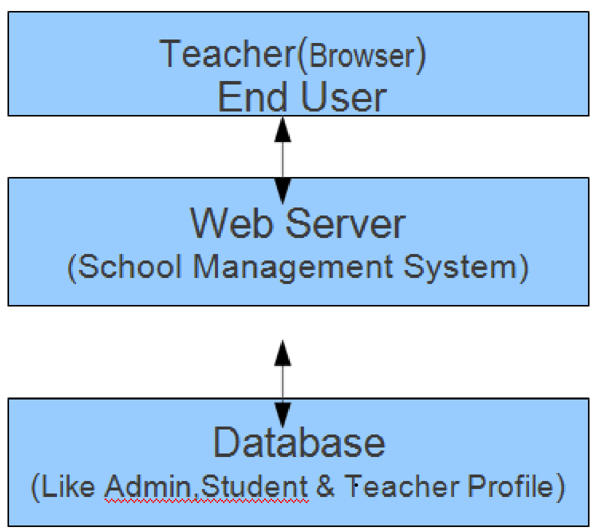


Figure 1.5 Overview

The Overall Description

The student management system allows authorized members to access the records of academically registered students. It can be used in various educational institutes across the globe and simplifies working of institutes

2.1 Product Perspective

The proposed system shall be developed using client/server architecture and be compatible with Microsoft Windows Operating System. The front end of the system will be developed using Dreamweaver and backend will be developed using MS SQL Server 2000.

2.1.1 Operations

i. The ONSMS will have following user-friendly and menu driven interfaces

ii. **Login:** to allow the entry of only authorized users through valid login Id and password.

iii. School Details: to maintain school details.

iv. Program Details: to maintain program details.

v. Scheme Details : to maintain scheme details of a school

vi. Paper Details: to maintain paper details of a scheme for a particular program

vii. Faculty Details: to maintain the faculty details.

2.1.2 Site Adaptation Requirements

2.2 Product Functions

The OSMS will allow access only to authorized users with specific roles (System administrator, Faculty and Teacher). Depending upon the user's role, he/she will be able to access only specific modules of the system.

A summary of major functions that the URS will perform

- A login facility for enabling only authorized access to the system.

- System administrator will be able to add, modify or delete program, school, scheme, paper and login information.
- Students will be able to add/modify his/her details and register for papers to be studied in the current semester.
- System administrator/Faculty will be able to generate reports.

2.3 User Characteristics

- Qualification: At least matriculation and comfortable with English.
- Experience: Should be well versed/informed about the registration process of the university.
- Technical Experience: Elementary knowledge of computers

2.4 General Constraints

There will only be one administrator.

- ·The delete operation is available only to the administrator. To reduce the complexity of the system, there is no check on delete operation. Hence, administrator should be very careful before deletion of any record and he/she will be responsible for data consistency.

2.5 Assumptions and Dependencies

- The login Id and password must be created by system administrator and communicated to the concerned user confidentially to avoid unauthorized access to the system.
- It is assumed that a student registering for the subsequent semester has been promoted to that semester by the university as per rules and has paid desired university fee.
- Registration process will be open only for specific duration

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 System Interfaces

In the very first step the manager will authenticate his/herself. Then main page will open where all categories of available staff and students are in Recycler View of Home. User After login , add in to staff and checking out of a confirmation pop up message will appear.

Admin

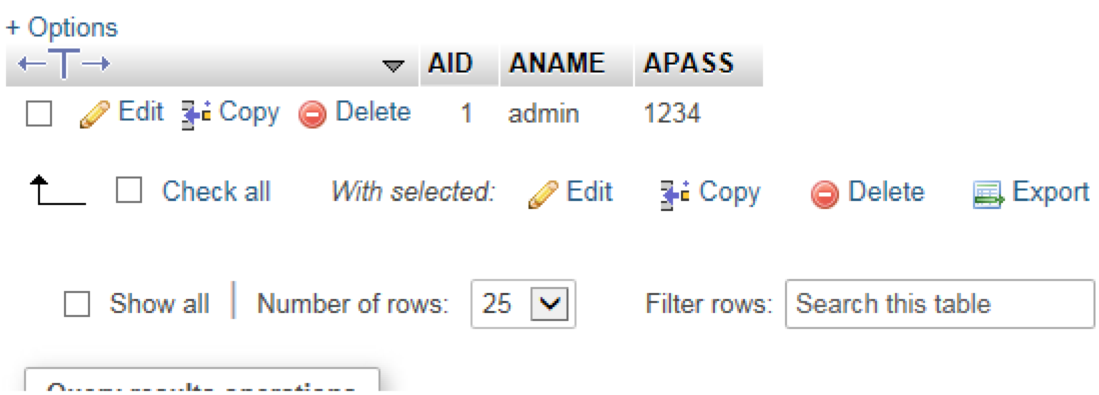


Figure 3.1.1. 3.1.1.1 Admin

Class handle

+ Options

	HID	TID	CLA	SEC	SUB
Edit Copy Delete	1	1	I	A	English
Edit Copy Delete	2	1	I	A	Math
Edit Copy Delete	3	1	VIII	B	English

Check all *With selected:* Edit Copy Delete Export

Figure 3.1.1.2 Class handle

Class database

+ Options

	CID	CNAME	CSEC
Edit Copy Delete	1	VIII	B
Edit Copy Delete	2	I	A
Edit Copy Delete	4	II	E
Edit Copy Delete	5	IX	F
Edit Copy Delete	6	X	A
Edit Copy Delete	7	V	A
Edit Copy Delete	8	VII	A
Edit Copy Delete	9	V	-

Check all *With selected:* Edit Copy Delete Export

Figure 3.1.1.3 Class database

Subject database

+ Options

				SID	SNAME
☐	Edit	Copy	Delete	1	Math
☐	Edit	Copy	Delete	2	English
☐	Edit	Copy	Delete	3	Data structure
☐	Edit	Copy	Delete	4	Compiler
☐	Edit	Copy	Delete	5	Visual Basic
☐	Edit	Copy	Delete	6	Networking
☐	Edit	Copy	Delete	7	psychology
☐	Edit	Copy	Delete	8	database management
☐	Edit	Copy	Delete	9	software Engineering

Figure 3.1.1.4 Subject database

Exam database

+ Options

				EID	ENAME	ETYPE	EDATE	SESSION	CLASS	SUB
☐	Edit	Copy	Delete	3	matric	II-Term	28-06-2019	FN	X	Compiler
☐	Edit	Copy	Delete	4	matric	I-Term	15-10-2019	FN	II	Networking
☐	Edit	Copy	Delete	5	middle	III-Term	13-05-2019	FN	V	Visual Basic
☐	Edit	Copy	Delete	6	seven	I-Term	15-11-2019	FN	X	software Engineering
☐	Edit	Copy	Delete	7	6th	II-Term	12-05-2019	AN	V	Visual Basic

☐ Check all *With selected:* Edit Copy Delete Export

Figure 3.1.1.5 Exam database

Marks database

+ Options

	MID	REGNO	SUB	MARK	TERM	CLASS
Edit Copy Delete	1	S102	Math	88	I-Term	I
Edit Copy Delete	2	s105	Compiler	78	II-Term	IX

Check all With selected: Edit Copy Delete Export

Figure 3.1.1.6 Marks database

Staff database

+ Options

	TID	TNAME	TPASS	QUAL	SAL	PNO	MAIL	PADDR	IMG
Edit Copy Delete	1	M WAQAS	1234	M.Sc Math	35000	03458712	wwKHURAwM@GMAIL.COM	wsrfyns7ki8ks	staff/Penguins.jpg
Edit Copy Delete	2	ALI	1234	BA	25000				
Edit Copy Delete	3	rauf	1234	BA	50000	03458712	wwKHURAwM@GMAIL.COM	4ftrgy4gy5	staff/Koala.jpg
Edit Copy Delete	4	khalid	1234	bs	10000				

Figure 3.1.1.7 Staff database

Student database

+ Options

	ID	RNO	NAME	FNAME	DOB	GEN	PHO	MAIL	ADDR	SCLASS	SSEC	SIMG
Edit Copy Delete	2	S102	Erum	Sharif	16-10-2002	Female	3068462112	bwpgojra@gmail.com	Gojra	I	A	student/The School-unif for-students th...
Edit Copy Delete	3	S103	Aslam	Ashraf	18-01-2002	Male	3068462118	waqagojra786@gmail.com	Bahawal Pur	I	A	student/3.jp
Edit Copy Delete	4	S104	junaid	ahmed	12-07-2005	Male	0345871259	khuram.shahzad564@gmail.com	wrerfawgasgaga	VIII	A	student/Trai Pinouts2.gif
Edit Copy Delete	5	S105	Hafiz khurram	Hafiz abdul sattar	6-09-2001	Male	0345871259	Khuram.shahzad564@gmail.com	House No. 707 Mohallah islampura Bahawalpur	IX	A	student/IMG 20190315-WA0021.jpg
Edit Copy Delete	7	S106	aeysha	Hafiz abdul sattar	13-08-2019	Male	03458712594	Khuram.shahzad564@gmail.com	rtrttretrew	I	B	student/IMG 20190315-WA0020.jpg

Figure 3.1.1.8 Student database

3.1.2 Interfaces

- First of all screen shows Sign In(For account holder) and Sign Up(For new User) buttons then move to food page.
- As described above food page will consists of information and photographs about all the available categories of foods and after clicking on done button a confirmation pop up message will appear.
- When admin/user give ID and password and click the login button the main page will open and login form will disappear.

3.1.3 Hardware Interfaces

a) Screen resolution of at least 640 x 480 or above.

b) Computer systems will be in the networked environment as it is a multi-user system.

3.1.4 Software Interfaces

a) MS-Windows Operating System

b) Microsoft Dreamweaver for designing front-end

c) MS SQL Server 2000 for backend

d) PLATEFORM : PHP LANGUAGE

e) INTEGRATED DEVELOPMENT ENVIRONMENT(IDE):ECLIPSE

3.1.5 Communications Interfaces

- None

3.2 Functional Requirements

- Our system provides these facilities.
- Store add staff and student detail in database.
- Our system provides a feature to update database in future as required.

- The system saves the record of the staff, student, exam and results.
- Admin must provide ID and Password before interacting with the system for authorization purpose.
- System must provide security from unauthorized access.

3.3 Use Cases

3.3.1 Use Case

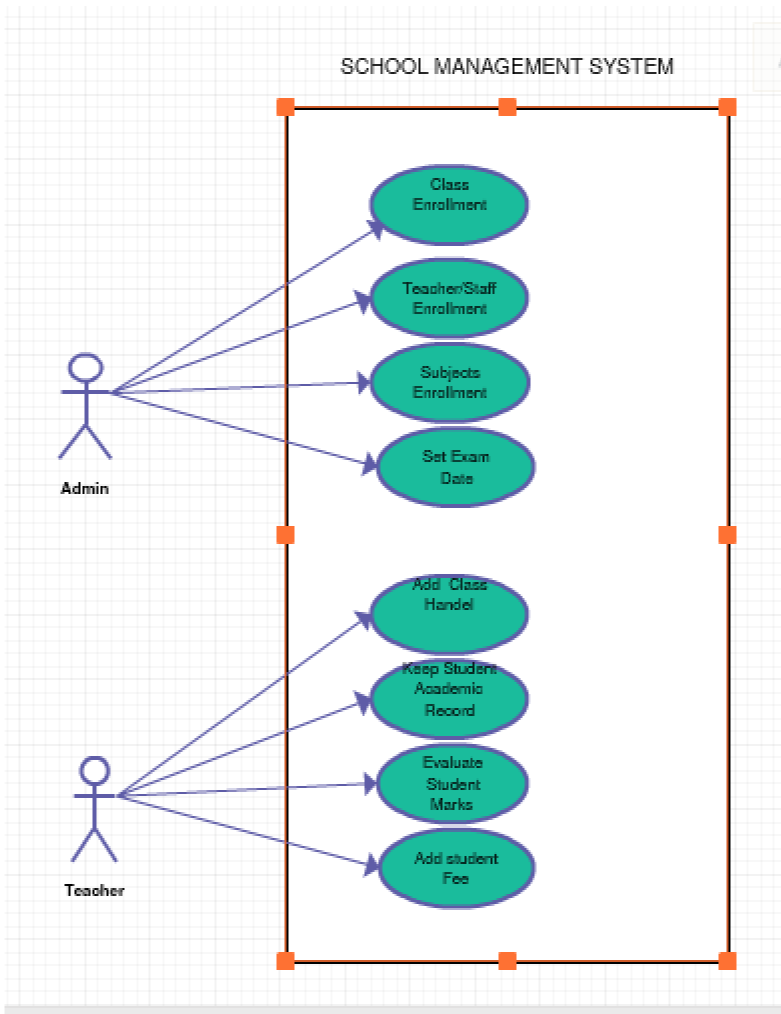
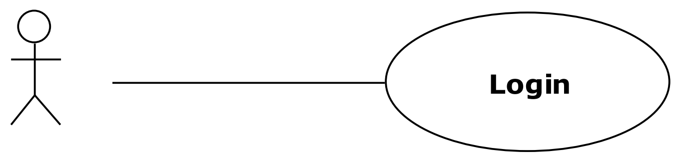
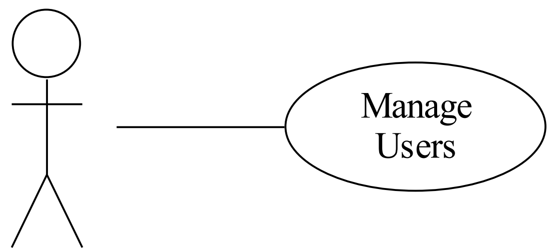


Figure 3.3.1.1 Use Case School Management System

Use case 2.

Use Case ID#1	UC-01
Use Case Name	**Login**
Actors	Administrator/User
Use Case Diagram	
Description	This use case describes the process of login verification for the users of the system
Pre-Condition	Database should be connected.
Primary Scenario	Step 1. The user will provide username and password Step 2. The user will click login. Step 3. The system will validate the login information. Step 4. The system will show the success message. Step 5. The system will open the main application.
Secondary Scenario	1. The administrator can click exit to close the application.
Exceptions	Step 2. Username/Password not supplied. Step 3. Username/Password is not valid. Step 5. Unable to connect to database, the application will be closed.
Post-Condition	The system shall apply the proper access rights defined for the current user.

Use Case ID	UC-02
Use Case Name	**Manage Users**
Actors	Administrator
Use Case Diagram	
Description	Administrator can Save, Update, View or Delete user information.
Pre-Condition	Administrator must be logged in.
Primary Scenario	Step 1. Administrator will click on view User information. Step 2:System will display User information form Step 3. The administrator would browse for the existing information of user. Step 4. The system will display its preview. Step 5. The administrator will save, update, delete user information Step 6. The system will take action according to select operation. Step 7. The administrator will click save. Step 3. The system will save the user information and display success message. Step 4. The Administrator will select click update. Step 5. The system will display all the information of the user.

	Step 6. The administrator will change the information and clicks save to update the user information. Step 7. The system will save the user information and display success message. Step 8. The administrator will click delete button. Step 9. The system will delete the selected user information and displays success message.
Secondary Scenario	1. The administrator can click exit to leave the process. 2. The administrator can come back to browse another user information.
Exceptions	Step 10. (a) Information not supplied. (b) Information not valid. Step 11. Invalid data entry in fields do not perform selected operation successfully.

3.4.1 Classes / Objects

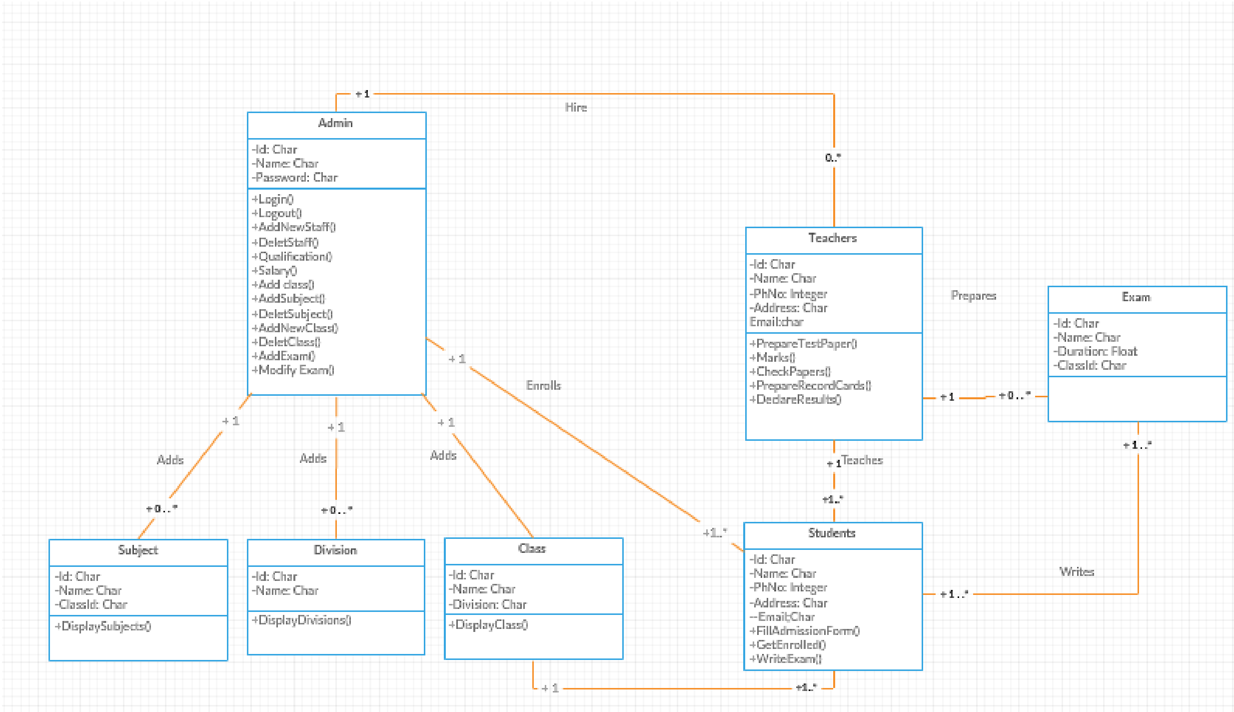


Figure 3.4.1.1 Classes / Objects

3.4.2 Class Object 2

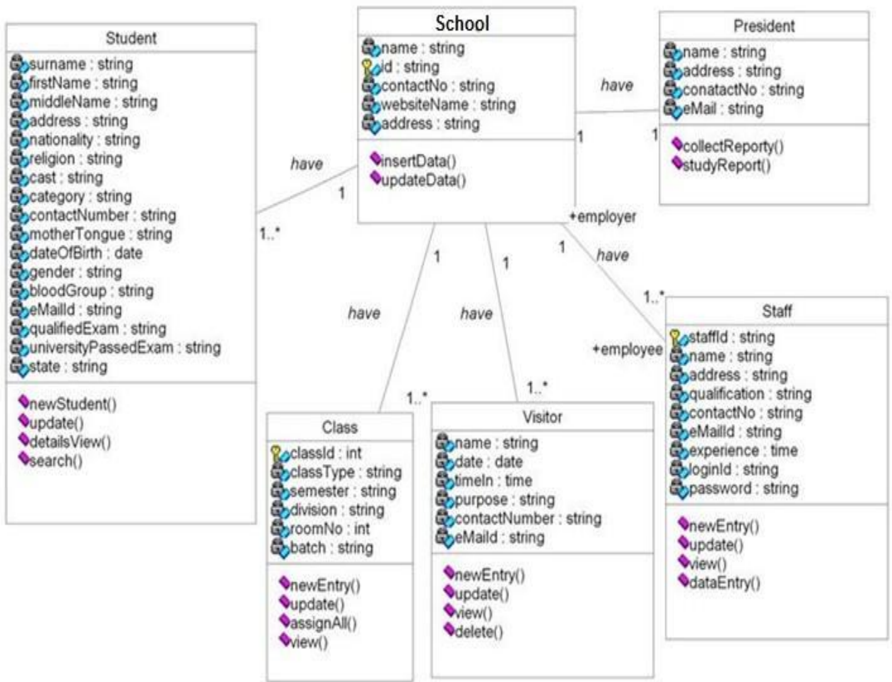


Figure 3.4.2.1 Class Object 2

- This section contains major categories of the “Taste It”.

a) **<Admin>**

b) **ID, Password**

c) **for login for interaction with system**

d) **<User>**

e) **Name,Cell number Mail**

f) **for storing information of user**

g) **<Exam>**

h) **Category for storing semester session categories**

3.5 Non-Functional Requirements.

3.5.1 Performance: Working will be fast.

3.5.2 Reliability: System will be easy to operate.

3.5.3 Availability: The system is easy to use because GUI is provided.

3.5.4 Security: System will restrict unauthorized access.

3.5.5 Maintainability: System will be easy to update as needed.

3.5.6 Portability: System will be easy to use on any Desktop Computer.

3.6 Inverse Requirements

- class only be selected from our given categories.

3.7 Logical Database Requirements

- Database must be valid for the system.
- The database must possess all the record of our system.
- Teacher is not allowed to interfere with the database.
- There will be different tables in database Admin authentication, user records and Student category record.

3.8 Design Constraints

- All fields must be filled correctly
- User's data will automatically be stored in firebase database on user sign up.
- After completing an operation display a message of record save or error.
- All necessary fields must be filled.

3.8.1 Standards Compliance

- Perform a system safety or security assessment
- Determine a target system failure rate

- Use the system target failure rate to determine the appropriate level of development rigor
- Use a formal requirements capture process
- Create software that adheres to an appropriate coding standard
- Trace all code back to their source requirements
- Develop all software and system test cases based on requirements
- Trace test cases to requirements
- Use coverage analysis to assess test completeness against both requirements and code
- For certification, collect and collate the process artifacts required to demonstrate that an appropriate level of rigor has been maintained.